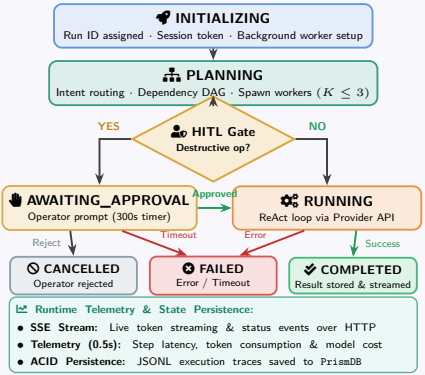
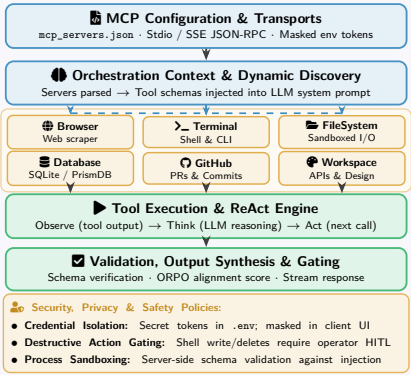
08. Agent Lifecycle, MCP Integration & Runtime Flow in PrismSpace

Runtime Orchestration & Tool Architecture: State-machine lifecycle with Human-in-the-Loop (HITL) safety gates, bi-directional MCP tool discovery, and real-time Server-Sent Events (SSE) telemetry.

Agent Lifecycle & State Flow



MCP Tool Integration & Runtime Flow



Core Infrastructure: **FastAPI Async Engine** (`hive_api.py`) | **Next.js 14 Frontend** (`AgentSwarm.tsx`) | **Dexie PrismDB** (ACID Storage) | **Multi-LLM Routing** (Groq, NVIDIA, OpenAI)